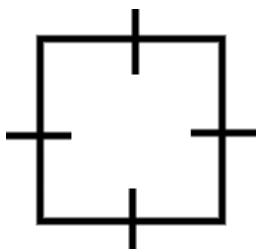




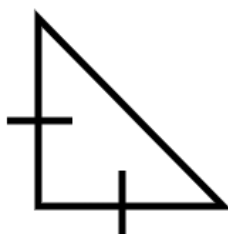
Area of a trapezium

Task A: Building an area

1) State how many of each shape should be used to create a composite shape with the stated area.



Area = 16 cm^2



Area = 8 cm^2



Area = 18 cm^2

a) Area = 24 cm^2

E.g.
You can use three
triangles.
 $3 \times 8 = 24$

b) Area = 50 cm^2

E.g.
You can use two squares
and one parallelogram.
 $16 + 16 + 18 = 50$

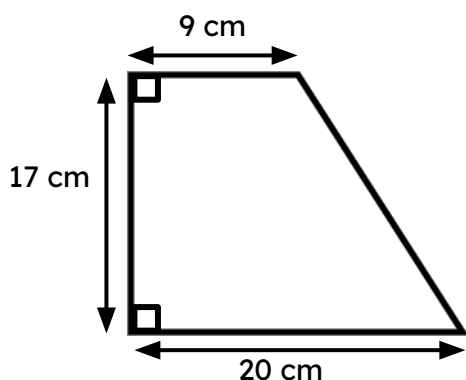
c) Area = 60 cm^2

E.g.
You can use one square, one
triangle and two parallelograms.
 $16 + 8 + 2 \times 18 = 60$

Task B: Area of a trapezium via composite shapes

1) Calculate the area of each trapezium.

a)



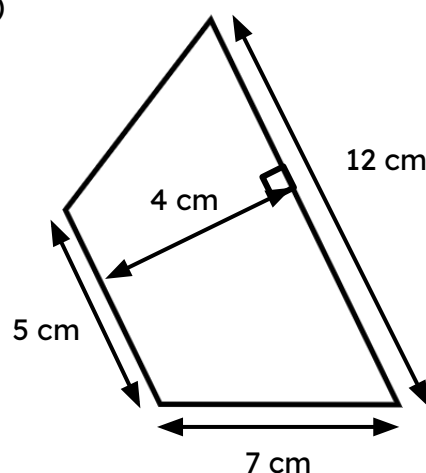
$$\text{Area} = 9 \times 17 + (20 - 9) \times 17 \times \frac{1}{2}$$

$$\text{Area} = 153 + 11 \times 17 \times \frac{1}{2}$$

$$\text{Area} = 153 + 93.5$$

$$\text{Area} = 246.5 \text{ cm}^2$$

b)



$$\text{Area} = 5 \times 4 + (12 - 5) \times 4 \times \frac{1}{2}$$

$$\text{Area} = 20 + 7 \times 4 \times \frac{1}{2}$$

$$\text{Area} = 20 + 14$$

$$\text{Area} = 34 \text{ cm}^2$$



Task C: Formula for the area of any trapezium**1) Instructions:**

- Take a piece of paper and fold it in half.
- Draw a **trapezium** on your piece of folded paper (try to make it different from mine).
- Cut out your **trapezium**. You should have two identical **trapeziums** due to your folded paper.
- Rotate one **trapezium** by half a turn. Place the two **trapeziums** next to each other.

Which quadrilateral did you make?

You have made a parallelogram

